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***B.Tech. Degree III Semester Supplementary Examination
April 2018***

**ME 15 – 1302 ELECTRICAL TECHNOLOGY
(2015 Scheme)**

Time : 3 Hours

Maximum Marks : 60

PART A(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Sketch the mechanical and electrical characteristics of DC series generator.
 (b) Explain the necessity of starters in DC motors.
 (c) Explain any 2 speed control method of DC shunt motor.
 (d) Derive EMF equation of transformer and show that emf/turn is same for both primary and secondary windings.
 (e) Explain the working principle of an autotransformer.
 (f) A potential transformer rated 2000/120 V and a current transformer rated 80/4 A is used to measure the voltage and current in a transmission line. If the voltmeter indicates 108 V and the ammeter indicates 1.5 A, calculate the voltage and current in the line.
 (g) Derive an expression for the pitch factor of an alternator.
 (h) If the induced emf in the stator of an 8 pole induction motor has a frequency of 50 Hz and that in the rotor is 1.5 Hz, at what speed is the motor running and what is the slip?
 (i) What are the factors affecting corona?
 (j) What are the advantages of high voltage transmission? State its limitations.

PART B

(4 × 10 = 40)

- II. (a) A 220 V shunt motor has armature and field resistance of 0.2 Ω and 220 Ω respectively. The motor is driving a constant load torque and running at 1000 rpm drawing 10 A current from the supply. Calculate the new speed and armature current if an external armature resistance of value 5 Ω is inserted in the armature circuit. Neglect armature reaction and saturation. (6)
 (b) Enumerate and classify the losses in a DC shunt motor. (4)
- OR**
- III. (a) Define commutation. Explain the process of commutation in dc generator with neat sketches. (5)
 (b) A 200 V, d.c. shunt machine has an armature resistance of 0.5 Ω and field resistance of 200 Ω . The machine is running at 1000 rpm as a motor drawing 31 A from the supply mains. Calculate the speed at which the machine must be driven to achieve this as generator. (5)

(P.T.O.)

- IV. The following data were obtained from testing a 48 kVA, 4800/240V, 50 Hz transformer. (10)

O.C test (from LV side):	240 V	2A	120 W
SC test (from HV side):	150 V	10A	600 W

- (i) Draw the equivalent circuit referred to the HV side and insert all the parameter values.
- (ii) At what kVA the transformer should be operated for maximum efficiency? Also calculate the value of maximum efficiency for 0.8 lagging factor at half load.

OR

- V. (a) Draw and explain the vector diagram of practical transformer when it is connected to an inductive load. (6)
- (b) A 20 kVA, 220 V/110 V, 50 Hz single phase transformer has full load copper loss = 200 W and core loss = 112.5 W. At what kVA and load power factor should the transformer be operated for maximum efficiency? (4)

- VI. (a) What is meant by synchronous condenser? Explain. (4)
- (b) A 3 Φ star connected alternator is rated at 1200 kVA, 11 kV. Under short circuit it takes 55A field current to produce full load current. The emf on open circuit for the same excitation is 1580 V/phase. Determine the synchronous reactance per phase and voltage regulation for (i) 0.8 p.f. lagging (ii) 0.8 p.f. leading. Neglect armature resistance. (6)

OR

- VII. (a) Derive the condition for maximum running torque of 3 ϕ induction motor. (4)
- (b) An 8 pole, 3 phase, 50 Hz induction motor runs at a speed of 800 rpm with an input power of 30 kW. The stator copper loss at this condition is 1000 W while the rotational losses are 500 W. find (i) Rotor Cu loss (ii) gross torque developed (iii) gross mechanical power developed (iv) net torque and mechanical power output. (6)

- VIII. (a) With a neat sketch explain nuclear power station. (6)
- (b) What is skin effect? (4)

OR

- IX. (a) What are the electrical equipments in power station? Explain. (6)
- (b) What is meant by load dispatching? (4)
